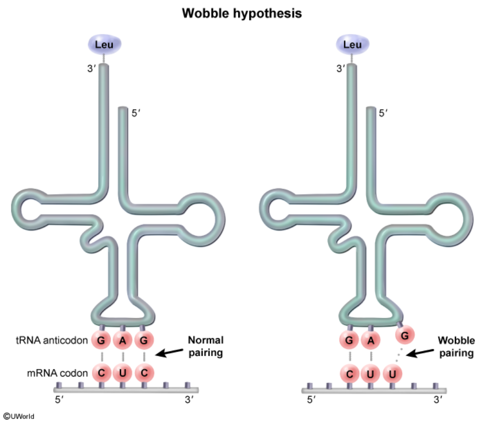
The genetic code is said to be degenerate because there are 64 different codons, but translation produces only 20 unique amino acids. The degeneracy of the genetic code is due to which mechanism?

- A. Exclusion of protein coding regions from mature mRNA [10%]
- B. Errors during tRNA charging [2%]
- C. Ambiguity of the tRNA for the amino acid [28%]
- ✔ D. Nontraditional base pairing of the anticodon with the third base of the codon [58%]

Correct
58% answered correctly

Explanation:



The genetic code explains how four kinds of DNA nucleotides (A, T, G, and C) can specify the 20 amino acids necessary for protein synthesis. Sequences of three nucleotides in DNA or RNA are known as codons. Genes contain codons that are sequentially transcribed into mRNA codons that code for amino acids during translation. Mathematically, there can only be 64 (or 4^3) possible arrangements of the four DNA nucleotides into codons. However, the genetic code is often considered *degenerate* or *redundant* because although there are 64 codons, only 20 amino acids are used for protein synthesis.

In translation, each mRNA codon (except the stop codons) base pairs with the complementary anticodon of a tRNA molecule that is "charged" with a corresponding amino acid. The anticodon of a single tRNA molecule can base pair with different mRNA codons that code for the same amino acid. This redundancy of the genetic code can be explained in part by the wobble hypothesis. This hypothesis states that the first two nucleotides on the mRNA codon require traditional (Watson-Crick) base pairing with their complementary nucleotides on the tRNA anticodon, but the third nucleotide on the codon may undergo less stringent ("wobble") base pairing in a non-Watson-Crick manner. Codon degeneracy is therefore explained by **nontraditional base pairing** at the **third position** of the codon and anticodon.

(Choice A) Alternative splicing cannot explain degeneracy because it refers to the post-transcriptional selection of exons for translation. Depending on the sets of exons selected, the same gene can encode different proteins.

(Choice B) tRNA charging refers to the covalent bonding (attachment) of an amino acid to the 3′-OH end of the tRNA molecule. *Errors* during tRNA charging do not explain genetic code degeneracy but can result in the wrong amino acid being added to the growing polypeptide chain (missense mutation).

(Choice C) The genetic code is redundant but not ambiguous. A single tRNA molecule can bind multiple different codons; however, a single tRNA can be charged with only one amino acid. Therefore, there is no ambiguity between a tRNA molecule and the amino acid it is meant to carry.

Educational objective:

The genetic code is considered "degenerate" because more than one codon can code for the same amino acid. This occurs because the third position of the mRNA codon and tRNA anticodon can undergo nontraditional base pairing, allowing a single tRNA molecule to bind different codons.

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